For the use of a Registered Medical Practitioner Only

PRESCRIBING INFORMATION

SIZONORM

(Quetiapine Tablets USP 25mg, 100mg, 200mg)

COMPOSITION

SIZONORM 25
Quetiapine tablet USP 25mg

Each film coated tablet contains:
Quetiapine fumarate 28.78 mg
Equivalent to quetiapine ....25 mg USP

SIZONORM 100
Quetiapine tablet USP 100mg

Each film coated tablet contains:
Quetiapine fumarate 115.12mg
Equivalent to quetiapine ....100 mg USP

SIZONORM 200
Quetiapine tablet USP 200mg

Each film coated tablet contains:
Quetiapine fumarate 230.24mg
Equivalent to quetiapine ....200 mg USP

Excipients: Silicified microcrystalline cellulose, Crospovidone, Colloidal anhydrous silica, Anhydrous citric acid, Polysorbate 80, Silicon dioxide, Magnesium stearate, Opadry II 33F58617 white, Opadry II 33F84535 Pink, Opadry II 33F82557 yellow, Purified water.

DESCRIPTION

SIZONORM 25
Peach, round, biconvex, film coated tablets plain on both sides.

SIZONORM 100
Yellow, round, biconvex, film coated tablets plain on one side and Break-line on other side.

The scored line is just for identification purpose and not meant to be halved.

SIZONORM 200
White, round, biconvex, film coated tablets plain on one side and Break-line on other side.

The scored line is just for identification purpose and not meant to be halved.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Antipsychotics, ATC code: N05A H04

Mechanism of Action
Quetiapine is an atypical antipsychotic agent. Quetiapine and the active human plasma metabolite, norquetiapine interact with a broad range of neurotransmitter receptors. Quetiapine and norquetiapine exhibit affinity for brain serotonin ($5HT_2$) and dopamine D_1- and D_2-receptors. It is this combination of receptor antagonism with a higher selectivity for $5HT_2$ relative to D_2-receptors, which is believed to contribute to the clinical antipsychotic properties and low extrapyramidal side effect (EPS) liability of quetiapine.

Quetiapine has no affinity for the norepinephrine transporter (NET) and low affinity for the serotonin $5HT_{1A}$ receptor, whereas norquetiapine has high affinity for both. Inhibition of NET and partial agonist action at $5HT_{1A}$ sites by norquetiapine may contribute to Quetiapine's therapeutic efficacy as an antidepressant. Quetiapine and norquetiapine have high affinity at histaminergic and adrenergic alpha 1 receptors and moderate affinity at adrenergic alpha 2 receptors. Quetiapine also has low or no affinity for muscarinic receptors, while norquetiapine has moderate to high affinity for several muscarinic receptor subtypes which may explain anti-cholinergic (muscarinic) effects.

Pharmacodynamic effects

Quetiapine is active in tests for antipsychotic activity, such as conditioned avoidance. It also blocks the action of dopamine agonists, measured either behaviourally or electrophysiologically, and elevates dopamine metabolite concentrations, a neurochemical index of D_2-receptor blockade.

Quetiapine is unlike standard antipsychotics and has an atypical profile. Quetiapine does not produce dopamine D_2 receptor supersensitivity after chronic administration. Quetiapine produces only weak catalepsy at effective dopamine D_2-receptor blocking doses. Quetiapine demonstrates selectivity for the limbic system by producing depolarisation blockade of the mesolimbic but not the nigrostriatal dopamine-containing neurones following chronic administration. Quetiapine exhibits minimal dystonic liability in haloperidol-sensitised or drug-naive Cebus monkeys after acute and chronic administration.

Pharmacokinetics

<u>Absorption</u>
Quetiapine is well absorbed and extensively metabolised following oral administration. The bioavailability of quetiapine is not significantly affected by administration with food. Steady-state peak molar concentrations of the active metabolite norquetiapine have been reported to be 35% of that reported for quetiapine.

The pharmacokinetics of quetiapine and norquetiapine have been reported to be linear across the approved dosing range.

Distribution
Quetiapine have been reported to be approximately 83% bound to plasma proteins.

Biotransformation
Quetiapine is extensively metabolised by the liver, with parent compound accounting for less than 5% of unchanged drug-related material in the urine or faeces, following the administration of radiolabelled quetiapine. CYP3A4 has been reported to be the primary enzyme responsible for cytochrome P450 mediated metabolism of quetiapine. Norquetiapine is primarily formed and eliminated via CYP3A4.

Approximately 73% of the radioactivity have been reported to be excreted in the urine and 21% in the faeces.

Quetiapine and several of its metabolites (including norquetiapine) have been reported to be weak inhibitors of human cytochrome P450 1A2, 2C9, 2C19, 2D6 and 3A4 activities. CYP inhibition have been reported only at concentrations approximately 5 to 50 fold higher than those reported at a dose range of 300 to 800 mg/day in humans. It is unlikely that co-administration of quetiapine with other drugs will result in clinically significant drug inhibition of cytochrome P450 mediated metabolism of the other drug. Quetiapine can induce cytochrome P450 enzymes. No increase in the cytochrome P450 activity have been reported after administration of quetiapine in psychotic patients.

Elimination
The elimination half-lives of quetiapine and norquetiapine have been reported to be approximately 7 and 12 hours, respectively. The average molar dose fraction of free quetiapine and the active human plasma metabolite norquetiapine has been reported to be <5% excreted in the urine.

Special populations

Gender
The kinetics of quetiapine do not differ between men and women.

Elderly
The mean clearance of quetiapine in the elderly have been reported to be approximately 30 to 50% lower than that reported in adults aged 18 to 65 years.

Renal impairment
The mean plasma clearance of quetiapine have been reported to reduce by approximately 25% in subjects with severe renal impairment (creatinine clearance less than 30 ml/min/1.73 m^2), but the individual clearance values have been reported to be within the range for normal subjects.

Hepatic impairment
The mean quetiapine plasma clearance have been reported to decrease with approx. 25% in persons with known hepatic impairment (stable alcohol cirrhosis). As quetiapine is extensively metabolised by the liver, elevated plasma levels are expected in the population with hepatic impairment. Dose adjustments may be necessary in these patients.

Paediatric population

At steady-state, the dose-normalised plasma levels of the parent compound, quetiapine 400 mg twice daily, in children and adolescents (10-17 years of age) have been reported to be in general similar to adults, though C_{max} in children has been reported at the higher end of the range reported in adults. The AUC and C_{max} for the active metabolite, norquetiapine, have been reported to be higher, approximately 62% and 49% in children (10-12 years), respectively and 28% and 14% in adolescents (13-17 years), respectively, compared to adults.

INDICATIONS

- Treatment of schizophrenia
- Treatment of acute manic episodes associated with bipolar I disorder, as either monotherapy or adjunct to lithium or divalproex.
- Treatment of depressive episodes associated with bipolar disorder.
- Maintenance treatment of bipolar I disorder as adjunct therapy to lithium or divalproex.

  The physician who elects to use quetiapine tor extended periods in Bipolar Disorder should periodically re-evaluate the long-term risks and benefits of the drug for the individual patient.

  Special Considerations in Treating Pediatric Schizophrenia and Bipolar 1 Disorder

  Paediatric schizophrenia and bipolar I disorder are serious mental disorders, however, diagnosis can be challenging. For paediatric schizophrenia, symptom profiles can be variable, and for bipolar I disorder, patients may have variable patterns of periodicity of manic or mixed symptoms. It is recommended that medication therapy for paediatric schizophrenia and bipolar I disorder be initiated only after a thorough diagnostic evaluation has been performed and careful consideration given to the risks associated with medication treatment. Medication treatment for both paediatric schizophrenia and bipolar I disorder is indicated as part of a total treatment program that often includes psychological, educational and social interventions.

- Prevention of recurrence in patients with bipolar disorder, in patients whose manic or depressive episode has responded for quetiapine treatment.

DOSE AND METHOD OF ADMINISTRATION

The scored line in the tablet is just for identification purpose and not meant to be halved.

Schizophrenia

Adults

SIZONORM should be administered twice daily, with or without food.

Quetiapine should generally be administered with an initial dose of 25 mg bid, with increases in increments of 25-50 mg bid or tid on the second and third day, as tolerated, to a target dose range of 300 to 400 mg daily by the fourth day, given bid or tid. Further dosage adjustments, if indicated, should generally reported at intervals of not less than 2 days, as steady state for quetiapine would not be achieved for approximately 1-2 days in the typical patient. When dosage adjustments are necessary, dose increments/decrements of 25-50 mg bid are recommended.

Adolescents (13-17 years)

Quetiapine should be administered twice daily. However, based on response and tolerability, quetiapine may be administered three times daily where needed.

The total daily dose for the initial five days of therapy is 50 mg (Day 1), 100 mg (Day 2), 200 mg (Day 3), 300 mg (Day 4) and 400 mg (Day 5). After day 5, the dose should be adjusted within the recommended dose range at 400 mg/day to 800 mg/day based on response and tolerability. Dosage adjustments should be in increments of no greater than 100 mg/day. Efficacy was reported with quetiapine at both 400 mg and 800 mg; however, no additional benefit was reported in the 800 mg group.

Maintenance Treatment

While there is no body of evidence available to answer the question of how long the patient treated with quetiapine should remain on it, the effectiveness of maintenance treatment is well established for many other antipsychotic drugs. It is recommended that responding patients be continued on quetiapine, but at the lowest dose needed to maintain remission. Patients should be periodically reassessed to determine the need for maintenance treatment.

Reinitiation of Treatment in Patients Previously Discontinued

Although there are no data reported to specifically address reinitiation of treatment, it is recommended that when restarting patients who have had an interval of less than one week off quetiapine, titration of quetiapine not required and the maintenance dose may be reinitiated. When restarting therapy of patients who have been off quetiapine for more than one week; the initial titration schedule should be followed.

Acute manic episodes associated with bipolar I disorder

Adults

Quetiapine should be administered twice daily, with or without food.

As monotherapy or as adjunct therapy to mood stabilizers (lithium or divalproex), the total daily dose for the first four days of therapy is 100 mg (Day 1), 200 mg (Day 2), 300 mg (Day 3) and 400 mg (Day 4). Further dosage adjustments up to 800 mg/day by Day 6 should be in increments of no greater than 200 mg/day.

The dose may be adjusted depending on clinical response and tolerability of the individual patient, within the range at 200 to 800 mg per day. The usual effective dose is in the range of 400 to 800 mg per day.

Children and Adolescents (10 to 17 years)

Quetiapine should be administered twice daily. However, based on response and tolerability quetiapine may be administered three times daily where needed.

The total daily dose for the initial five days of therapy is 50 mg (Day 1), 100 mg (Day 2), 200 mg (Day 3), 300 mg (Day 4) and 400 mg (Day 5). After day 5, the dose should be adjusted within the recommended dose range of 400 to 600 mg/day based on response and tolerability. Dosage adjustments should be in increments of no greater than 100 mg/day. Efficacy was reported with quetiapine at both 400 mg and 600 mg; however, no additional benefit was reported in the 600 mg group.

Depressive episodes associated with bipolar disorder
Quetiapine should be administered once daily at bedtime, with or without food.

Quetiapine should be titrated as follows: 50 mg (Day 1), 100mg (Day 2), 200 mg (Day 3) and 300 mg (Day 4). Quetiapine can be titrated to 400 mg on Day 5 and up to 600 mg by Day 8. Antidepressant efficacy was reported with quetiapine at 300 mg and 600 mg however no additional benefit was reported in the 600 mg group.

Maintenance treatment of bipolar I disorder as adjunct therapy to lithium or divalproex.

Adults
Generally, in the maintenance phase, patients continued on the same dose on which they were stabilized during the stabilization phase.

Children and Adolescents (10 to 17 years)
The effectiveness of quetiapine for longer than 3 weeks has not been reported in children and adolescents. While there is no body of evidence available to answer the question of how long the patient treated with quetiapine should be maintained, it is generally recommended that responding patients be continued beyond the acute response, but at the lowest dose needed to maintain remission. Patients should be periodically reassessed to determine the need for maintenance treatment.

Preventing recurrence in bipolar disorder
For preventing recurrence of manic, mixed or depressive episodes in bipolar disorder, patients who have responded to quetiapine for acute treatment of bipolar disorder should continue therapy at the same dose. The dose may be adjusted depending on clinical response and tolerability of the individual patient, within the range of 300 to 800 mg/day administered twice daily. It is important that the lowest effective dose is used for maintenance therapy.

Elderly
As with other antipsychotics, quetiapine should be used with caution in the elderly, especially during the initial dosing period. The rate of dose titration may need to be slower, and the daily therapeutic dose lower, than that used in younger patients, depending on the clinical response and tolerability of the individual patient. The mean plasma clearance of quetiapine was reported to be reduced in elderly subjects when compared to younger patients.

Efficacy and safety has not been reported in patients over 65 years with depressive episodes in the framework of bipolar disorder.

Pediatric Use
In general, the adverse reactions were reported in children and adolescents were similar to those in the adult population with few exceptions. Increases in systolic and diastolic blood pressure occurred in children and adolescents and did not occur in adults. Orthostatic hypotension was reported more frequently in adults compared to children and adolescents.

Schizophrenia
The efficacy and safety of quetiapine in the treatment of schizophrenia in adolescents aged 13 to 17 years were reported. Safety and effectiveness of quetiapine in paediatric patients less than 13 years of age with schizophrenia have not been reported.

Maintenance
The safety and effectiveness of quetiapine in the maintenance treatment of bipolar disorder has not been reported in pediatric patients <18 years of age. The safety and effectiveness of quetiapine in the maintenance treatment of schizophrenia has not been reported in any patient population, including paediatric patients.

Bipolar Mania
The efficacy and safety of quetiapine has been reported in the treatment of mania in children and adolescents ages 10 to 17 years with bipolar I disorder. Safety and effectiveness of quetiapine in paediatric patients less than 10 years of age with bipolar mania have not been reported.

Bipolar Depression
Safety and effectiveness of quetiapine in pediatric patients <18 years of age with bipolar depression have not been reported. Some differences in the pharmacokinetics of quetiapine were reported between children/adolescents (10 to 17 years of age) and adults. When adjusted for weight, the AUC and C_{max} of quetiapine were reported to be lower in children and adolescents compared to adults. The pharmacokinetics of the active metabolite, norquetiapine, were similar between children/adolescents and adults after adjusting for weight.

Renal impairment
Dosage adjustment is not necessary in patients with renal impairment.

Hepatic impairment
Quetiapine is extensively metabolised by the liver. Therefore, quetiapine should be used with caution in patients with known hepatic impairment, especially during the initial dosing period. Patients with known hepatic impairment should be started with 25 mg/day. The dosage should be increased daily with increments of 25-50 mg/day until an effective dosage, depending on the clinical response and tolerability of the individual patient.

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients of this product.
- Concomitant administration of cytochrome P450 3A4 inhibitors, such as HIV-protease inhibitors, azole-antifungal agents, erythromycin, clarithromycin and nefazodone, is contraindicated.

WARNINGS AND PRECAUTIONS

As quetiapine has several indications, the safety profile should be considered with respect to the individual patient's diagnosis and the dose being administered.

Suicide/suicidal thoughts or clinical worsening
Depression in bipolar disorder is associated with an increased risk of suicidal thoughts, self-harm and suicide (suicide-related events). This risk persists until significant remission occurs. As improvement may not occur during the first few weeks or more of treatment, patients should be closely monitored until such improvement occurs. It is general clinical experience that the risk of suicide may increase in the early stages of recovery.

In addition, physicians should consider the potential risk of suicide-related events after abrupt cessation of quetiapine treatment, due to the known risk factors for the disease being treated. Other psychiatric conditions for which quetiapine is prescribed can also be associated with an increased risk of suicide related events. In addition, these conditions may be co-morbid with major depressive episodes. The same precautions observed when treating patients with major depressive episodes should therefore be observed when treating patients with other psychiatric disorders.

Patients with a history of suicide related events, or those exhibiting a significant degree of suicidal ideation prior to commencement of treatment are known to be at greater risk of suicidal thoughts or suicide attempts, and should receive careful monitoring during treatment. An increased risk of suicidal behaviour has been reported with antidepressants drugs in patients less than 25 years old.

Close supervision of patients and in particular those at high risk should accompany drug therapy especially in early treatment and following dose changes. Patients (and caregivers of patients) should be alerted about the need to monitor for any clinical worsening, suicidal behaviour or thoughts and unusual changes in behaviour and to seek medical advice immediately if these symptoms present.

In patients with major depressive episodes in bipolar disorder, an increased risk of suicide-related events has been reported in young adult patients (younger than 25 years of age) who were treated with quetiapine.

Metabolic risk
Given the observed risk for worsening of their metabolic profile, including changes in weight, blood glucose (see hyperglycaemia) and lipids, patients' metabolic parameters should be assessed at the time of treatment initiation and changes in these parameters should be regularly controlled for during the course of treatment. Worsening in these parameters should be managed as clinically appropriate.

Extrapyramidal symptoms
Quetiapine has been reported to be associated with an increased incidence of extrapyramidal symptoms (EPS) in adult patients treated for major depressive episodes in bipolar disorder.

The use of quetiapine has been associated with the development of akathisia, characterised by a subjectively unpleasant or distressing restlessness and need to move often accompanied by an inability to sit or stand still. This is most likely to occur within the first few weeks of treatment. In patients who develop these symptoms, increasing the dose may be detrimental.

Tardive dyskinesia
Tardive dyskinesia is a syndrome of potentially irreversible, involuntary, dyskinetic movements that may develop in patients treated with antipsychotic drugs including quetiapine. If signs and symptoms of tardive dyskinesia appear, dose reduction or discontinuation of quetiapine should be considered. The symptoms of tardive dyskinesia can worsen or even arise after discontinuation of treatment.

Somnolence and dizziness
Quetiapine treatment has been associated with somnolence and related symptoms, such as sedation. For treatment of patients with bipolar depression, onset was usually reported within the first 3 days of treatment and was predominantly of mild to moderate intensity. Patients experiencing somnolence of severe intensity may require more frequent contact for a minimum of 2 weeks from onset of somnolence, or until symptoms improve and treatment discontinuation may need to be considered.
Orthostatic hypotension
Quetiapine treatment has been associated with orthostatic hypotension and related dizziness which, like somnolence has onset usually during the initial dose-titration period. This could increase the occurrence of accidental injury (fall), especially in the elderly population. Therefore, patients should be advised to exercise caution until they are familiar with the potential effects of the medication.

Quetiapine should be used with caution in patients with known cardiovascular disease, cerebrovascular disease, or other conditions predisposing to hypotension. Dose reduction or more gradual titration should be considered if orthostatic hypotension occurs, especially in patients with underlying cardiovascular disease.

Sleep apnoea syndrome
Sleep apnoea syndrome has been reported in patients using quetiapine. In patients receiving concomitant central nervous system depressants and who have a history of or are at risk for sleep apnoea, such as those who are overweight/obese or are male, quetiapine should be used with caution.

Cardiovascular
Quetiapine should be used with caution in patients with known cardiovascular disease, cerebrovascular disease, or other conditions predisposing to hypotension. Quetiapine may induce orthostatic hypotension, especially during the initial dose-titration period and therefore dose reduction or more gradual titration should be considered if this occurs.

Seizures
No data is reported about the incidence of seizures in patients with a history of seizure disorder. As with other antipsychotics, caution is recommended when treating patients with a history of seizures.

Neuroleptic malignant syndrome
Neuroleptic malignant syndrome has been associated with antipsychotic treatment, including quetiapine. Clinical manifestations include hyperthermia, altered mental status, muscular rigidity, autonomic instability, and increased creatine phosphokinase. In such an event, quetiapine should be discontinued and appropriate medical treatment given.

Severe neutropenia and agranulocytosis
Severe neutropenia (neutrophil count $<0.5 \times 10^9$/L) has been reported with quetiapine. Most cases of severe neutropenia have been reported within a couple of months of starting therapy with quetiapine. There was no apparent dose relationship. Some fatal cases have been reported. Possible risk factors for neutropenia include pre-existing low white blood cell count (WBC) and history of drug induced neutropenia. However, some cases have been reported in patients without pre-existing risk factors. Quetiapine should be discontinued in patients with a neutrophil count $<1.0 \times 10^9$/L. Patients should be observed for signs and symptoms of infection and neutrophil counts followed (until they exceed 1.5×10^9/L)).

Neutropenia should be considered in patients presenting with infection or fever, particularly in the absence of obvious predisposing factor(s), and should be managed as clinically appropriate.

Patients should be advised to immediately report the appearance of signs/symptoms consistent with agranulocytosis or infection (e.g. fever, weakness, lethargy, or sore throat) at any time during quetiapine therapy. Such patients should have a WBC count and an absolute neutrophil count (ANC) performed promptly, especially in the absence of predisposing factors.

Anti-cholinergic (muscarinic) effects

Norquetiapine, an active metabolite of quetiapine, has moderate to strong affinity for several muscarinic receptor subtypes. This contributes to ADRs reflecting anti-cholinergic effects when quetiapine is used at recommended doses, when used concomitantly with other medications having anti-cholinergic effects, and in the setting of overdose. Quetiapine should be used with caution in patients receiving medications having anti-cholinergic (muscarinic) effects. Quetiapine should be used with caution in patients with a current diagnosis or prior history of urinary retention, clinically significant prostatic hypertrophy, intestinal obstruction or related conditions, increased intraocular pressure or narrow angle glaucoma.

Interactions

Concomitant use of quetiapine with a strong hepatic enzyme inducer such as carbamazepine or phenytoin substantially decreases quetiapine plasma concentrations, which could affect the efficacy of quetiapine therapy. In patients receiving a hepatic enzyme inducer, initiation of quetiapine treatment should only occur if the physician considers that the benefits of quetiapine outweigh the risks of removing the hepatic enzyme inducer. It is important that any change in the inducer is gradual, and if required, replaced with a non-inducer (e.g. sodium valproate).

Weight

Weight gain has been reported in patients who have been treated with quetiapine, and should be monitored and managed as clinically appropriate as in accordance with utilised antipsychotic guidelines.

Hyperglycemia and Diabetes mellitus

Hyperglycemia in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients treated with atypical antipsychotics. Assessment of the relationship between atypical antipsychotics use and glucose abnormalities is complicated by the possibility of an increased background risk of diabetes mellitus in patients with schizophrenia and the increasing incidence of diabetes mellitus in the general population. Given these confounders, the relationship between atypical antipsychotic use and hyperglycemia-related adverse events is not completely understood. However, epidemiological studies suggest an increased risk of treatment-emergent hyperglycemia-related adverse events in patients treated with the atypical antipsychotics. Precise risk estimates for hypergycemia-relaled adverse events in patients treated with atypical antipsychotics are not available.

Patients with an established diagnosis of diabetes mellitus who are started on atypical antipsychotics should be monitored regularly for worsening of glucose control. Patients with risk factors tor diabetes mellitus (e.g. obesity, family history of diabetes) who are starling treatment with atypical anti psychotics should undergo fasting blood glucose testing at the beginning of treatment and periodically during treatment. Any patient treated with atypical antipsychotics should be monitored for symptoms of hyperglycemia including polydipsia, polyuria, polyphagia, and weakness. Patients who develop symptoms of hyperglycemia during treatment with atypical antipsychotics should undergo fasting blood glucose testing. In some cases, hyperglycemia has resolved when the atypical antipsychotic was discontinued; however, some patients required continuation of anti-diabetic treatment despite discontinuation of the suspect drug.

Lipids

Increase in triglycerides, LDL and total cholesterol, and decreases in HDL cholesterol have been reported with quetiapine. Lipid changes should be managed as clinically appropriate.

QT prolongation

QT prolongation has been reported with quetiapine at the therapeutic doses and in overdose. As with other antipsychotics, caution should be exercised when quetiapine is prescribed in patients with cardiovascular disease or family history of QT prolongation. Also, caution should be exercised when quetiapine is prescribed either with medicines known to increase QT interval or with concomitant neuroleptics, especially in the elderly, in patients with congenital long QT syndrome, congestive heart failure, heart hypertrophy, hypokalaemia or hypomagnesaemia.

Cardiomyopathy and myocarditis

Cardiomyopathy and myocarditis have been reported with quetiapine, however, a causal relationship to quetiapine has not been established. Treatment with quetiapine should be reassessed in patients with suspected cardiomyopathy or myocarditis.

Withdrawal

Acute withdrawal symptoms such as insomnia, nausea, headache, diarrhoea, vomiting, dizziness and irritability have been reported after abrupt cessation of quetiapine. Gradual withdrawal over a period of at least one to two weeks is advisable.

Elderly patients with dementia-related psychosis

Quetiapine is not approved for the treatment of dementia-related psychosis.

An approximately 3-fold increased risk of cerebrovascular adverse events has been reported in the dementia population with some atypical antipsychotics. The mechanism for this increased risk is not known. An increased risk cannot be excluded for other antipsychotics or other patient populations. Quetiapine should be used with caution in patients with risk factors for stroke.

It has been reported that elderly patients with dementia-related psychosis are at an increased risk of death. The the incidence of mortality in quetiapine treated patients was higher. The patients in these reported studies died from a variety of causes that were consistent with expectations for this population. These data do not report a causal relationship between quetiapine treatment and death in elderly patients with dementia.

Dysphagia

Dysphagia and aspiration have been reported with quetiapine. Although a causal relationship with aspiration pneumonia has not been established, quetiapine should be used with caution in patients at risk for aspiration pneumonia.

Constipation and intestinal obstruction

Constipation represents a risk factor for intestinal obstruction. Constipation and intestinal obstruction have been reported with quetiapine. This includes fatal reports in patients who are at higher risk of intestinal obstruction, including those that are receiving multiple concomitant medications that decrease intestinal motility and/or may not report symptoms of constipation. Patients with intestinal obstruction/ileus should be managed with close monitoring and urgent care.

Venous thromboembolism (VTE)

Cases of venous thromboembolism (VTE) have been reported with antipsychotic drugs. Since patients treated with antipsychotics often present with acquired risk factors for VTE, all possible risk factors for VTE should be identified before and during treatment with quetiapine and preventive measures undertaken.

Pancreatitis

Pancreatitis has been reported with quetiapine. Among post marketing reports, while not all cases were confounded by risk factors, many patients had factors which are reported to be associated with pancreatitis such as increased triglycerides, gallstones, and alcohol consumption.

Paediatric population

Certain adverse events have reported at a higher frequency in children and adolescents compared to adults (increased appetite, elevations in serum prolactin, vomiting, rhinitis and syncope), or may have different implications for children and adolescents (extrapyramidal symptoms and irritability) and one was identified that has not been previously reported in adult (increases in blood pressure). Changes in thyroid function tests have also been reported in children and adolescents.

Furthermore, the long-term safety implications of treatment with quetiapine on growth and maturation have not been reported. Long-term implications for cognitive and behavioural development are not reported.

In children and adolescent patients, quetiapine was associated with an increased incidence of extrapyramidal symptoms (EPS) in patients treated for schizophrenia, bipolar mania and bipolar depression.

Additional information

Quetiapine data reported in combination with divalproex or lithium in acute moderate to severe manic episodes is limited; however, combination therapy was well tolerated.

Misuse and abuse

Cases of misuse and abuse have been reported. Caution may be needed when prescribing quetiapine to patients with a history of alcohol or drug abuse.

Effects on ability to drive and use machines

Given its primary central nervous system effects, quetiapine may interfere with activities requiring mental alertness. Therefore, patients should be advised not to drive or operate machinery, until individual susceptibility to this is reported.

DRUG INTERACTIONS

Given the primary central nervous system effects of quetiapine, quetiapine should be used with caution in combination with other centrally acting medicinal products and alcohol.

Caution should be exercised treating patients receiving other medications having anti-cholinergic (muscarinic) effects.

Cytochrome P450 (CYP) 3A4 is the enzyme that is primarily responsible for the cytochrome P450 mediated metabolism of quetiapine. Concomitant administration of quetiapine (dosage of 25 mg) with ketoconazole, a CYP3A4 inhibitor, has been reported to increase the AUC of quetiapine. On the basis of this, concomitant use of quetiapine with CYP3A4 inhibitors is contraindicated. It is also not recommended to consume grapefruit juice while on quetiapine therapy.

Co-administration of carbamazepine has been reported to be significantly increase the clearance of quetiapine. This increase in clearance reduces systemic quetiapine exposure; although a greater effect has been reported in some patients. As a consequence of this interaction, lower plasma concentrations can occur, which could affect the efficacy of quetiapine therapy. Co-administration of quetiapine and phenytoin (another microsomal enzyme inducer) caused a greatly increased clearance of quetiapine. In patients receiving a hepatic enzyme inducer, initiation of quetiapine treatment should only occur if the physician considers that the benefits of quetiapine outweigh the risks of removing the hepatic enzyme inducer. It is important that any change in the inducer is gradual, and if required, replaced with a non-inducer (e.g. sodium valproate).

The pharmacokinetics of quetiapine were not significantly altered by co-administration of the antidepressants imipramine (a known CYP 2D6 inhibitor) or fluoxetine (a known CYP 3A4 and CYP 2D6 inhibitor).

The pharmacokinetics of quetiapine were not significantly altered by co-administration of the antipsychotics risperidone or haloperidol. Concomitant use of quetiapine and thioridazine reported an increased clearance of quetiapine.

The pharmacokinetics of quetiapine were not altered following co-administration with cimetidine.

The pharmacokinetics of lithium were not altered when co-administered with quetiapine.

The pharmacokinetics of sodium valproate and quetiapine were not altered to a clinically relevant extent when co-administered.

Interaction with commonly used cardiovascular medicinal products have not been reported.

Caution should be exercised when quetiapine is used concomitantly with medicinal products reported to cause electrolyte imbalance or to increase QT interval.

There have been reports of false positive results in enzyme immunoassays for methadone and tricyclic antidepressants in patients who have taken quetiapine. Confirmation of questionable immunoassay screening results by an appropriate chromatographic technique is recommended.

UNDESIRABLE EFFECTS

The most commonly reported adverse drug reactions (ADRs) with quetiapine are somnolence, dizziness, headache, dry mouth, withdrawal (discontinuation) symptoms, elevations in serum triglyceride levels, elevations in total cholesterol (predominantly LDL cholesterol), decreases in HDL cholesterol, weight gain, decreased haemoglobin and extrapyramidal symptoms.

Table: ADRs associated with quetiapine therapy

The frequencies of adverse events are ranked according to the following: Very common, common, uncommon, rare, very rare and not known (cannot be estimated from the available data).

SOC	**Very Common**	**Common**	**Uncommon**	**Rare**	**Very Rare**	**Not known**
Blood and lymphatic system disorders	Decreased haemoglobin	Leucopenia decreased neutrophil count, eosinophils increased	Neutropenia, Thrombocytopenia, Anaemia, platelet count decreased [10]	Agranulocytosis		
Immune system disorders			Hypersensitivity (including allergic skin reactions)		Anaphylactic reaction	
Endocrine disorders		Hyperprolactinaemia [11], decreases in total T_4, decreases in free T_4, decreases in total T_3, increases in TSH	Decreases in free T_3, Hypothyroidism		Inappropriate antidiuretic hormone secretion	
Metabolism and nutritional disorders	Elevations in serum triglyceride levels [8], Elevations in total cholesterol (predominantly LDL cholesterol) [9], Decreases in HDL cholesterol [13] Weight gain [6]	Increased appetite, blood glucose increased to hyperglycaemic levels[4]	Hyponatraemia [14], Diabetes Mellitus Exacerbation of pre-existing diabetes	Metabolic syndrome		
Psychiatric disorders		Abnormal dreams and nightmares, Suicidal ideation and suicidal behaviour [15]		Somnambulism and related reactions such as sleep talking and sleep related eating disorder		
Nervous system disorders	Dizziness [3, 12], somnolence [1, 12] headache, Extrapyramidal symptoms,	Dysarthria	Seizure, Restless legs syndrome, Tardive dyskinesia, Syncope [12]			
Cardiac disorders		Tachycardia [3], Palpitations [16]	QT prolongation, Bradycardia			
Eye disorders		Vision blurred				
Vascular disorders		Orthostatic hypotension [3, 12]		Venous thromboembolism		
Respiratory, thoracic and mediastinal disorder		Dyspnoea [16] sleep apnoea*	Rhinitis			
Gastrointestinal disorders	Dry mouth	Constipation, dyspepsia, vomiting [17]	Dysphagia [5]	Pancreatitis, Intestinal obstruction/Ileus		
Hepato-biliary disorders		Elevations in serum alanine aminotransferase (ALT) [2], Elevations in gamma-GT levels [2]	Elevations in serum aspartate aminotransferase (AST)	Jaundice, Hepatitis		
Skin and					Angioedema,	Toxic

SOC	Very Common	Common	Uncommon	Rare	Very Rare	Not known
subcutaneous tissue disorders					Stevens-Johnson syndrome	Epidermal Necrolysis, Erythema Multiforme
Musculoskeletal and connective tissue disorders					Rhabdomyolysis	
Renal and urinary disorders			Urinary retention			
Pregnancy, puerperium and perinatal conditions						Drug withdrawal syndrome neonatal
Reproductive system and breast disorders			Sexual dysfunction	Priapism, galactorrhoea, breast swelling, menstrual disorder		
General disorders and administration site conditions	Withdrawal (discontinuation) symptoms [7]	Mild asthenia, peripheral oedema, irritability, pyrexia		Neuroleptic malignant syndrome, hypothermia		
Investigations				Elevations in blood creatine phosphokinase		

1. Somnolence may occur, usually during the first two weeks of treatment and generally resolves with the continued administration of quetiapine.
2. Asymptomatic elevations (shift from normal to >3 x ULN at any time) in serum transaminase (ALT, AST) or gamma-GT levels have been reported in some patients administered quetiapine. These elevations were usually reversible on continued quetiapine treatment.
3. As with other antipsychotics with alpha$_1$ adrenergic blocking activity, quetiapine may commonly induce orthostatic hypotension, associated with dizziness, tachycardia and, in some patients, syncope, especially during the initial dose-titration period.
4. Fasting blood glucose ≥126 mg/dL (≥7.0 mmol/L) or a non-fasting blood glucose ≥200 mg/dL (≥11.1 mmol/L) on at least one occasion.
5. An increase in the rate of dysphagia with quetiapine vs. placebo was only reported in the clinical studies in bipolar depression.
6. Based on >7% increase in body weight from baseline. Occurs predominantly during the early weeks of treatment in adults.
7. The following withdrawal symptoms have been reported most frequently: insomnia, nausea, headache, diarrhoea, vomiting, dizziness, and irritability. The incidence of these reactions had decreased significantly after 1 week post-discontinuation.
8. Triglycerides ≥200 mg/dL (≥2.258 mmol/L) (patients ≥18 years of age) or ≥150 mg/dL (≥1.694 mmol/L) (patients <18 years of age) on at least one occasion.
9. Cholesterol ≥240 mg/dL (≥6.2064 mmol/L) (patients ≥18 years of age) or ≥200 mg/dL (≥5.172 mmol/L) (patients <18 years of age) on at least one occasion. An increase in LDL cholesterol of ≥30 mg/dL (≥0.769 mmol/L) has been very commonly reported. Mean change among patients who had this increase was 41.7 mg/dL (≥1.07 mmol/L).
10. Platelets ≤100 x 10^9/L on at least one occasion.
11. Prolactin levels (patients >18 years of age): >20 µg/L (>869.56 pmol/L) males; >30 µg/L (>1304.34 pmol/L) females at any time.
12. May lead to falls.
13. HDL cholesterol: <40 mg/dL (1.025 mmol/L) males; <50 mg/dL (1.282 mmol/L) females at any time.
14. Shift from >132 mmol/L to ≤132 mmol/L on at least one occasion.
15. Cases of suicidal ideation and suicidal behaviours have been reported during quetiapine therapy or early after treatment discontinuation.
16. These reports often occurred in the setting of tachycardia, dizziness, orthostatic hypotension and/or underlying cardiac/respiratory disease.
17. Based upon the increased rate of vomiting in elderly patients (≥65 years of age).

* Atypical antipsychotic drugs, such as quetiapine, have been associated with cases of sleep apnoea, with or without concomitant weight gain. In patients who have history of or are at risk for sleep apnoea, **SIZONORM** should be prescribed with caution.

Cases of QT prolongation, ventricular arrhythmia, sudden unexplained death, cardiac arrest and *torsades de pointes* have been reported with the use of neuroleptics and are considered class effects. Severe cutaneous adverse reactions (SCARs), including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported in association with quetiapine treatment.

Extrapyramidal Symptoms

The extrapyramidal symptoms reported were akathisia, extrapyramidal disorder, tremor, dyskinesia, dystonia, restlessness, muscle contractions involuntary, psychomotor hyperactivity and muscle rigidity. Severe cutaneous adverse reactions (SCARs), including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported in association with quetiapine treatment.

Thyroid Levels

Quetiapine treatment was associated with dose-related decreases in thyroid hormone levels.

The changes in thyroid hormone levels are generally not associated with clinically symptomatic hypothyroidism. The reduction in total and free T_4 was maximal within the first six weeks of quetiapine treatment, with no further reduction during long-term treatment. In nearly all cases, cessation of quetiapine treatment was associated with a reversal of the effects on total and free T_4, irrespective of the duration of treatment. In eight patients, where TBG was measured, levels of TBG were unchanged.

Paediatric population

The same ADRs described above for adults should be considered for children and adolescents. The following table summarizes ADRs that reported in a higher frequency category in children and adolescent patients (10-17 years of age) than in the adult population or ADRs that have not been identified in the adult population.

Table: ADRs in children and adolescents associated with quetiapine therapy that occur in a higher frequency than adults, or not identified in the adult population

The frequencies of adverse events are ranked according to the following: Very common, common, uncommon, rare and very rare.

SOC	Very Common	Common
Endocrine disorders	Elevations in prolactin[1]	
Metabolism and nutritional disorders	Increased appetite	
Nervous system disorders	Extrapyramidal symptoms[2]	Syncope
Vascular disorders	Increases in blood pressure	
Respiratory, thoracic and mediastinal disorders		Rhinitis
Gastrointestinal disorders	Vomiting	
General disorders and administration site conditions		Irritability[2]

1. Prolactin levels (patients < 18 years of age) : >20 µg/L (>869.56 pmol/L) males; >26 µg/L (>1130.428 pmol/L) females at any time. Less than 1% of patients had an increase to a prolactin level >100 µg/L.
2. Note: The frequency is consistent to that reported in adults, but might be associated with different clinical implications in children and adolescents as compared to adults.

PREGNANCY AND LACTATION

The safety and efficacy of quetiapine during human pregnancy have not yet been established. Following some pregnancies in which quetiapine was used, neonatal withdrawal symptoms have been reported. Therefore, quetiapine should only be used during pregnancy if the benefits justify the potential risks.

There have been published reports of quetiapine excretion into human breast milk, however the degree of excretion was not consistent. Women who are breast feeding should therefore be advised to avoid breast feeding while taking quetiapine.

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation.

SIZONORM should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

OVERDOSE

Fatal outcome has been reported following an acute overdose at 13.6 g, and in post-marketing reports on doses as low as 6 g of quetiapine alone. However, survival has also been reported following acute overdoses of up to 30 g. In post-marketing experience, there have been very rare reports of overdose of quetiapine alone, resulting in death or coma or QT-prolongation.

Patients with pre-existing severe cardiovascular disease may be at an increased risk of the effects of overdose.

In general, reported signs and symptoms were those resulting an exaggeration of the drug's known pharmacological effects, i.e., drowsiness and sedation, tachycardia, hypotension and anticholinergic effects.

Management of overdose
There is no specific antidote to quetiapine. In cases of severe signs, the possibility of multiple drug involvement should be considered, and intensive care procedures are recommended, including establishing and maintaining a patent airway, ensuring adequate oxygenation and ventilation, and monitoring and support of the cardiovascular system. In this context, published reports in the selling of anti-cholinergic symptoms describe a reversal of severe CNS effects, including coma and delirium, with administration of intravenous physostigmine (1-2 mg), under continuous EGG monitoring. Whilst the prevention of absorption in overdose has not been investigated, gastric lavage (after intubation, if patient is unconscious) and administration of activated charcoal together with a laxative should be considered.

In cases of quetiapine overdose, refractory hypotension should be treated with appropriate measures such as intravenous fluids and/or sympathomimetic agents (epinephrine and dopamine should be avoided, since beta stimulation may worsen hypotension in the setting of quetiapine-induced alpha blockade).

Close medical supervision and monitoring should be continued until the patient recovers.

Pack Style: **SIZONORM Tablet** is available as 10 tablets is packed in Blister and this 03 blister packed in Show box along with pack insert. The scored line is just for identification purpose and not meant to be halved.

STORAGE

Store below 30^0 C.

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